MPLS Japan 2022

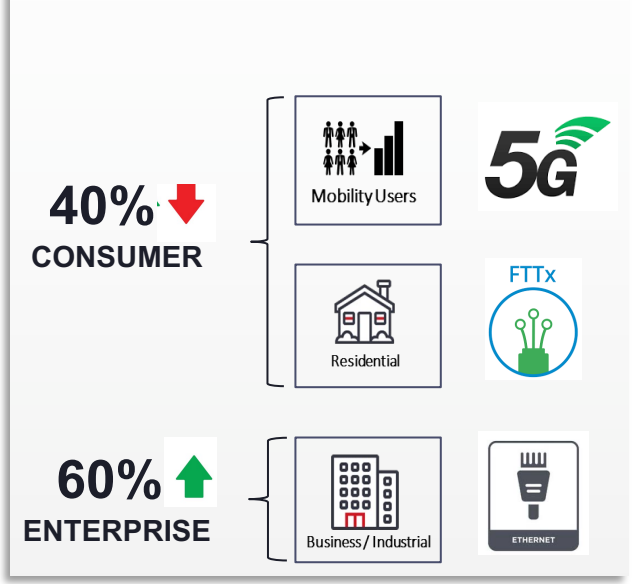
Coherent Modem evolution beyond 800G

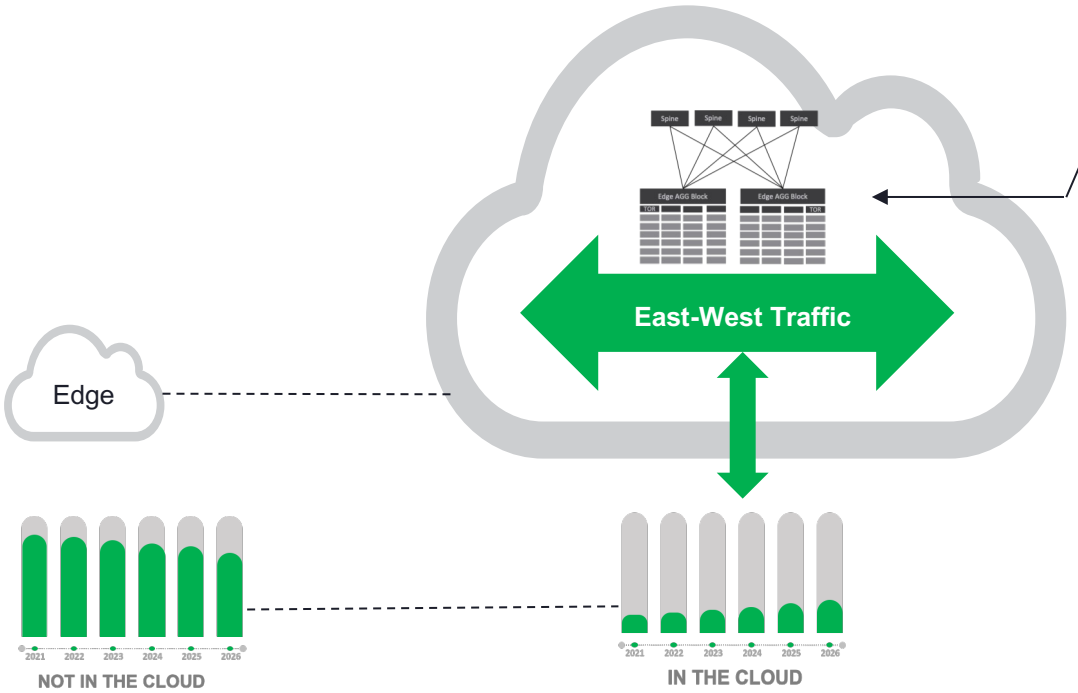
Gautam Billa
Vice President System Engineering @ Ciena International

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The Digital Data Ecosystem

- Total data creation (2x in 4Y)
- 80% Data still On Prem
- Small N-S > Large E-W
- 70% Global Traffic inside DC





1

Hyperscale DC Networking

- App Software – Containers
- Compute / Storage Evolution
- AI-ML / Data Lakes / Analytics
- Focus on Power / Footprint
- SerDes / Switch Evolution
- Concept of System Balance

2

Telco Networking (WAN)

- Traffic Flows / Patterns
- Resiliency & SLA based
- Inter-Op Requirements
- Customized Designs
- High Performance Optics

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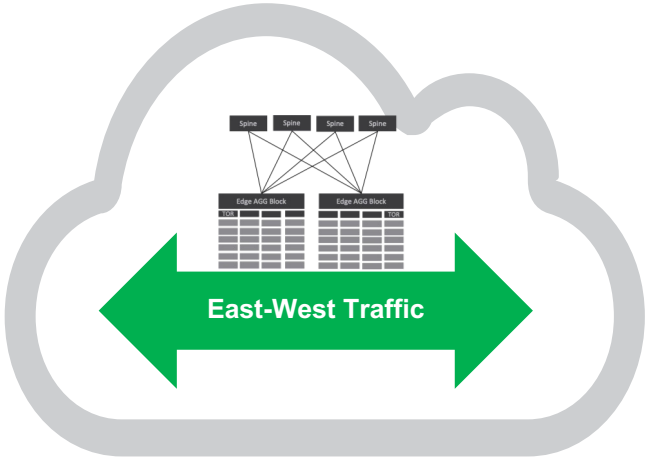
Hyperscale DC Networking

App Software – Containers

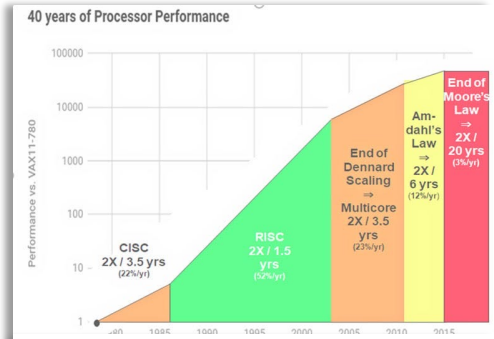
- Micro-service Architecture
- Distributed Computing
- Demand for Data movement

AI-ML / Data Lakes / Analytics

- High BW E-W data transfers
- Big Data Models / Algorithms
- Data Lakes / Fat pipe exchange



Compute / Storage Evolution

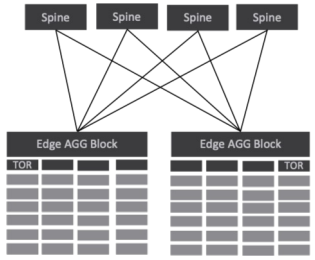


- Hardware Acceleration
- GPUs | NPUs | TPUs
- Fast I/O over ETH

Concept of System Balance

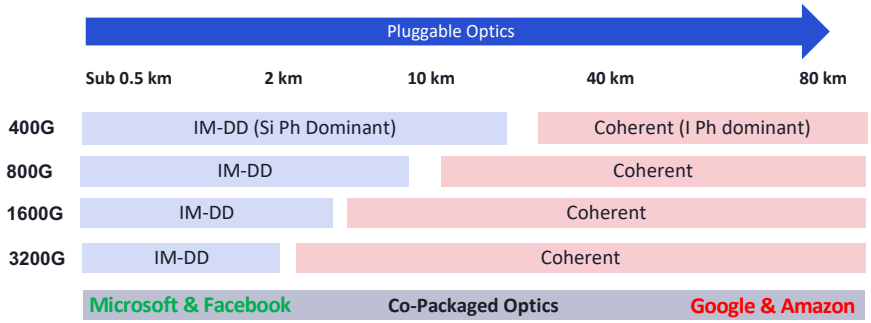
- Another server not the answer
- Efficient Utilization of Resources
- Non-Blocking Data Exchange

SerDes / Switch Evolution



Focus on Power / Footprint

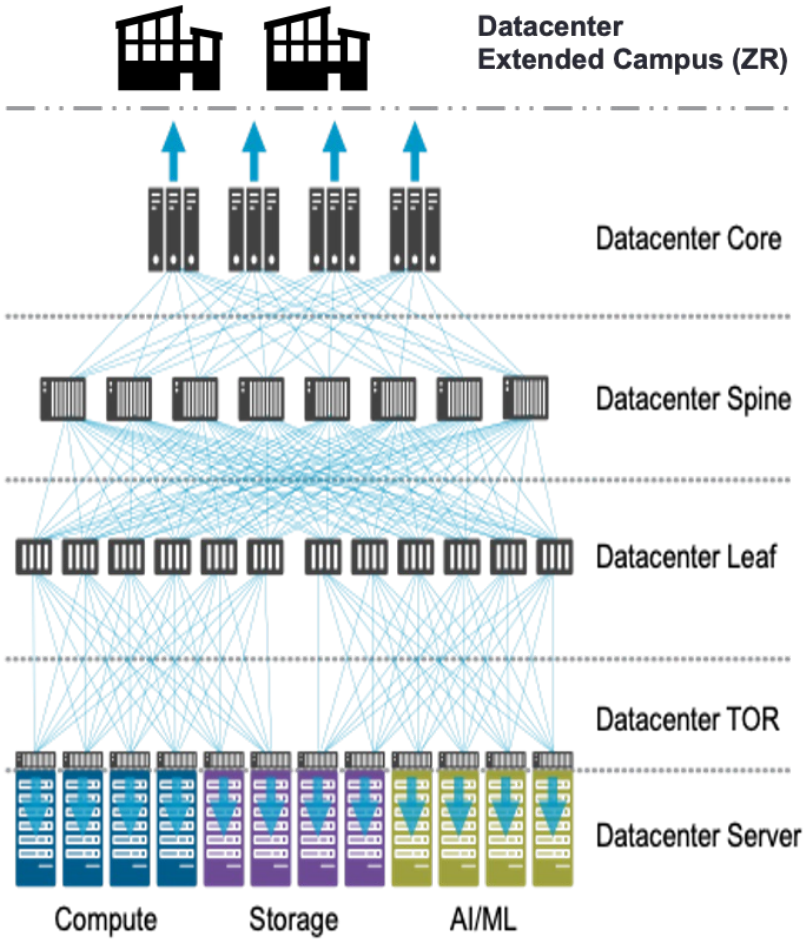
Lane Technology	Port Rate G	Ports per 1 RU	Switch Silicon T SERDES
4x100G	400	32	12.8
8x100G	800	32	25.6
16x100G 8x200G	1600	32	51.2
16x200G	3200	32	102.4

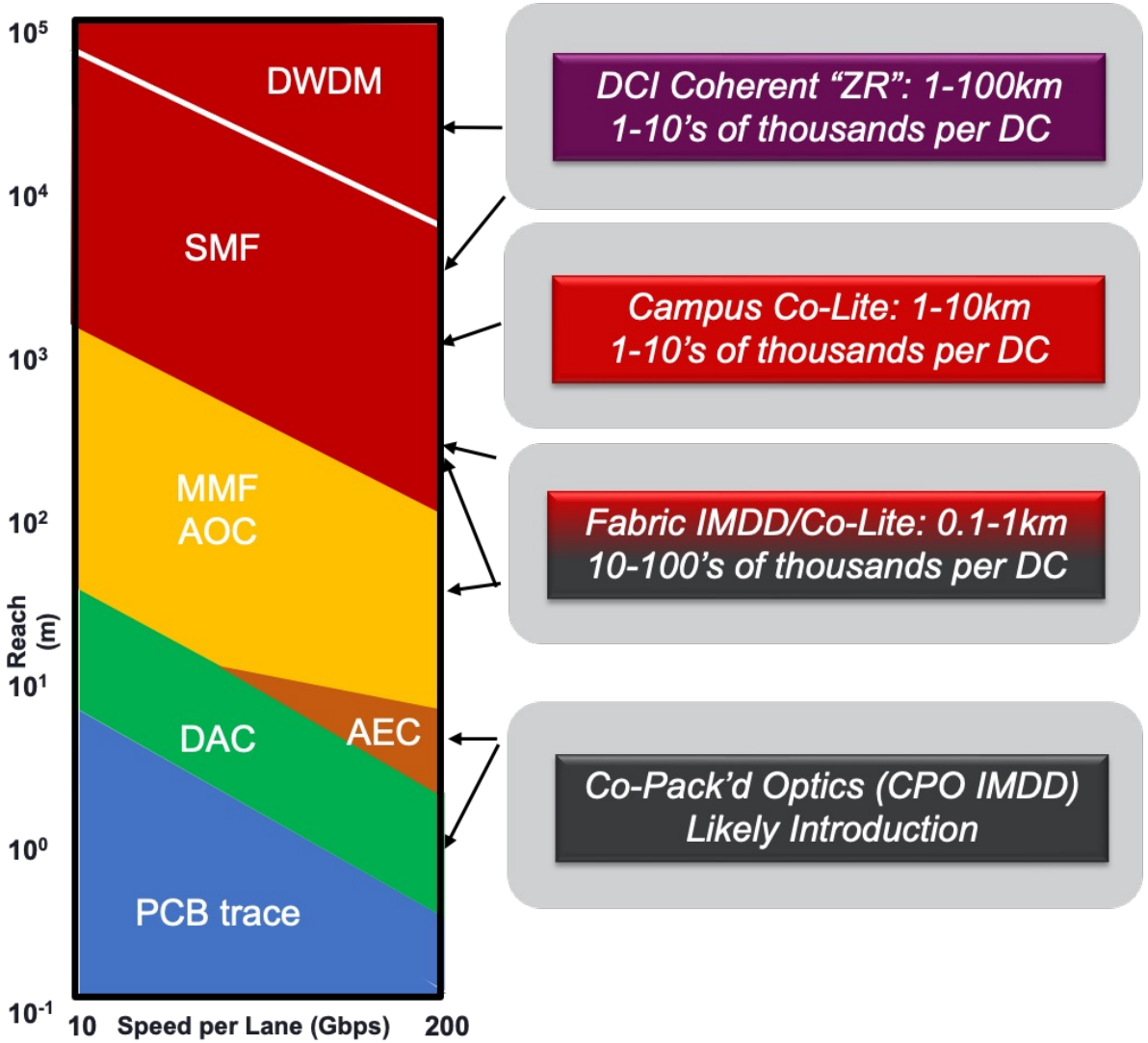


- Plugs Vs CPO – co-existence
- IMDD Vs Coherent Lite – co-existence
- The promise of “Carbon Neutral / Negative”
- Power / Performance / Cost balance of Optics

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Next Gen DC Networking

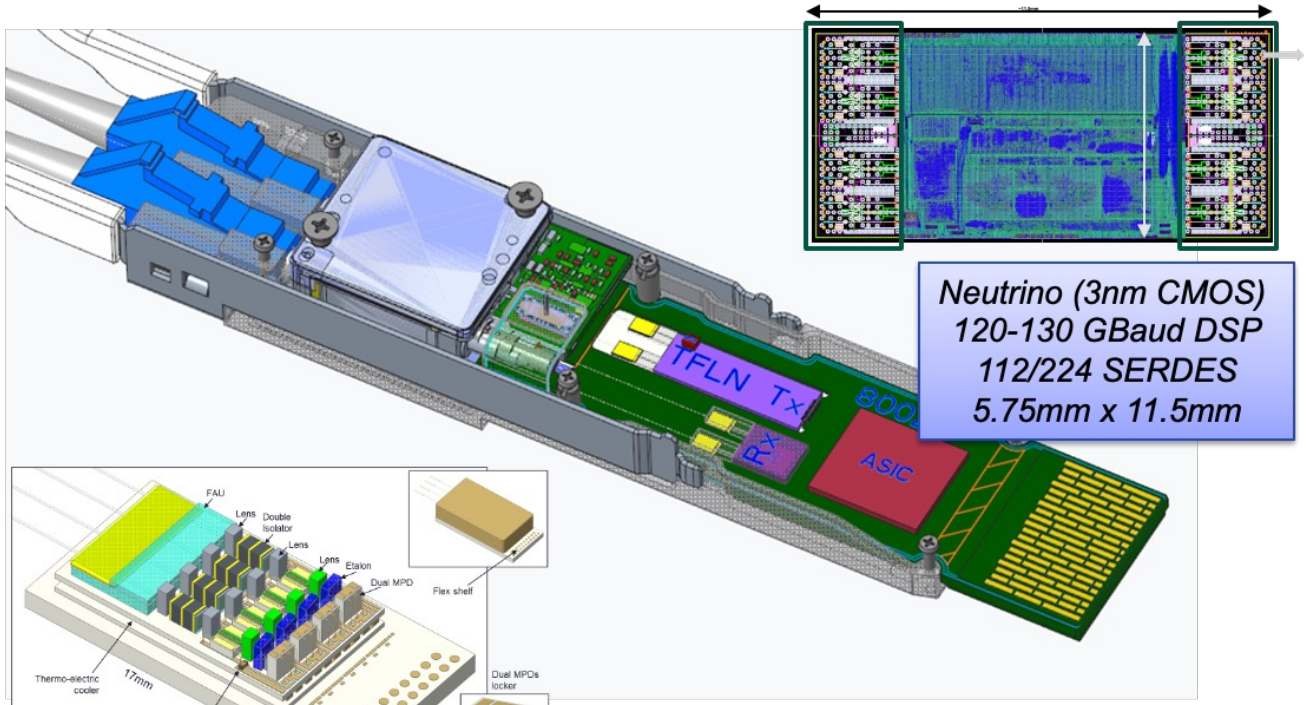




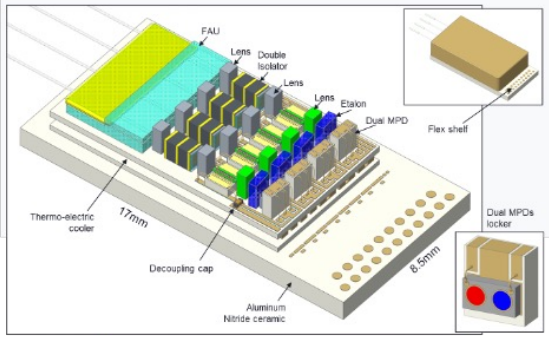
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Next Gen Pluggable Technology: 800ZR/ZR+ | 1.6T | 3.2T Campus

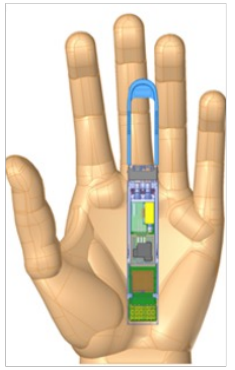
Leveraging High-Baud, Power Optimized DSP & SiPhotonics & Thin Film E/O Platform

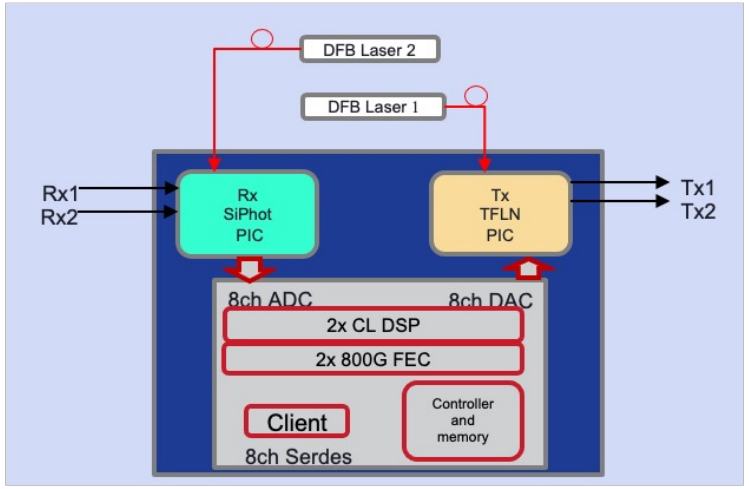


Neutrino (3nm CMOS)
120-130 GBaud DSP
112/224 SERDES
5.75mm x 11.5mm



2-4 λ Laser Array



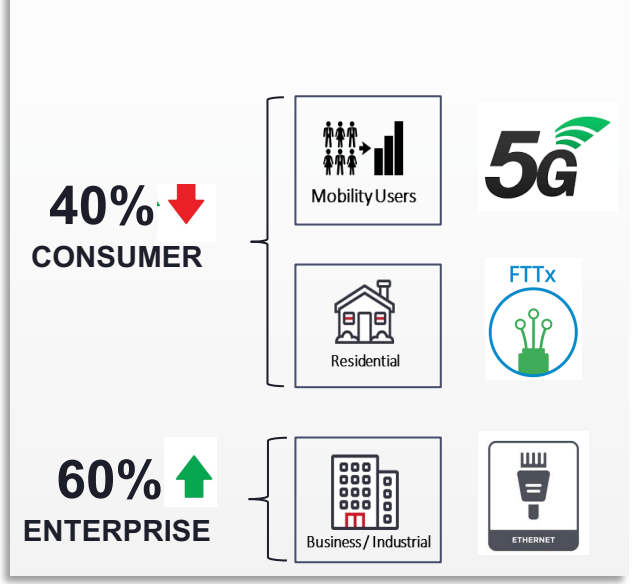


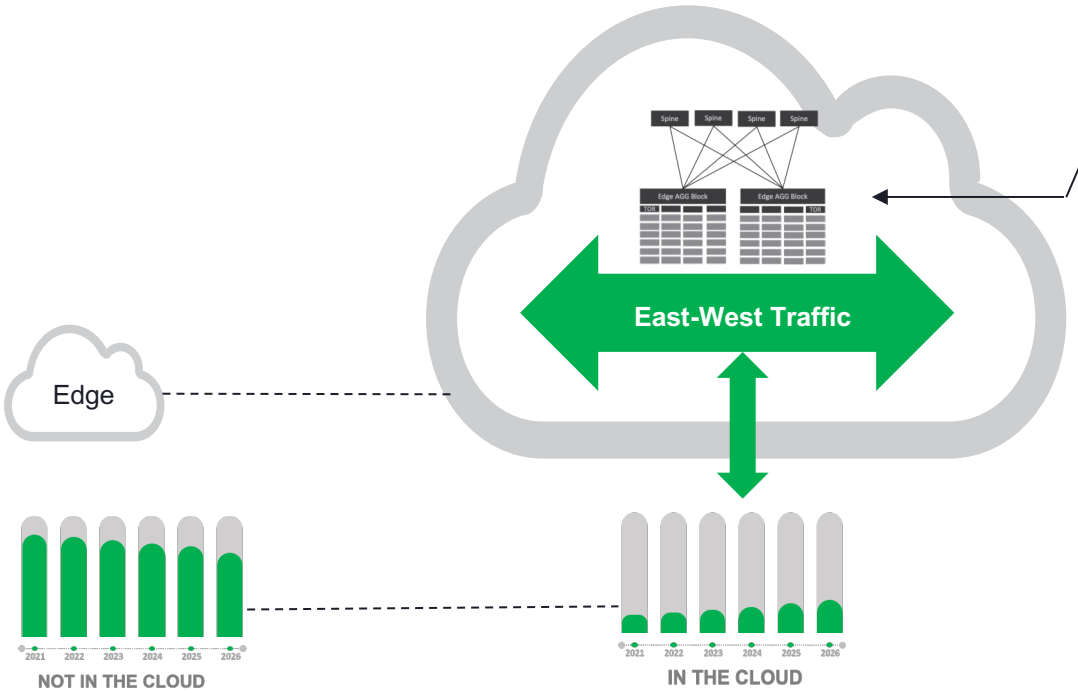
800Gb/s ZR at ~20W
1.6Tb/s Campus at <30W

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1 **Hyperscale DC Networking**

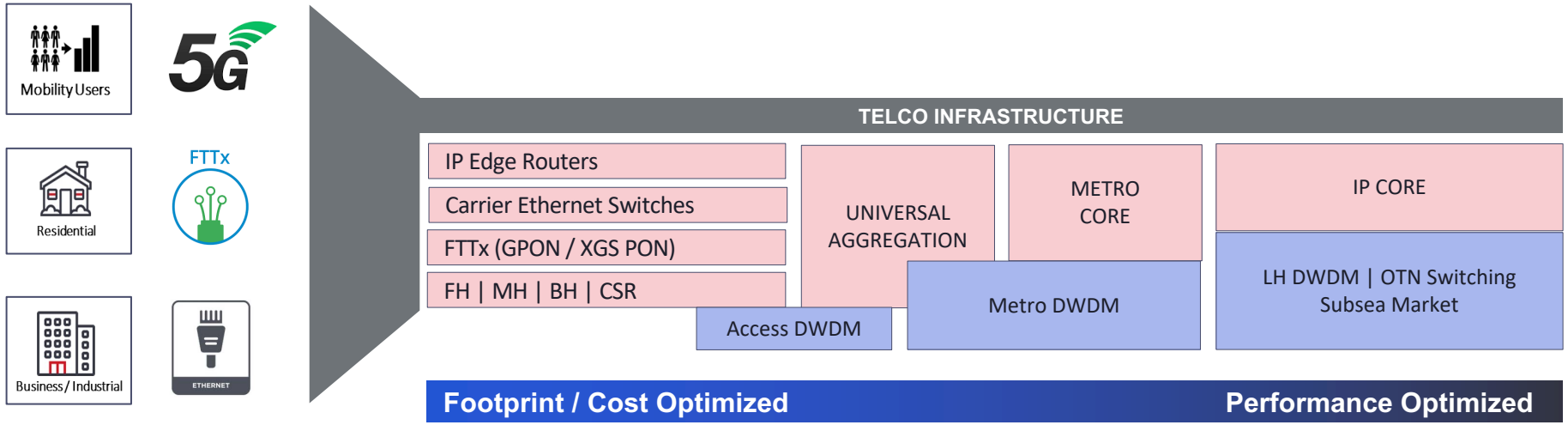
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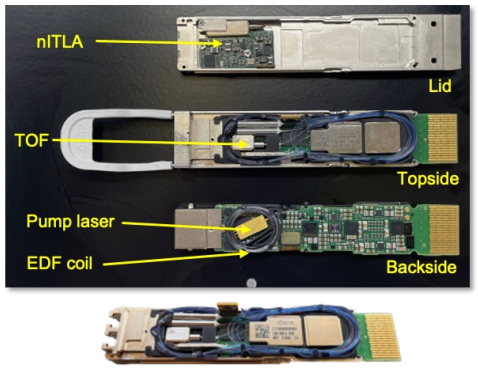
Telco Networking (WAN)



TODAY - GEN-5 TECHNOLOGY

400Gb/s DWDM evolution





800G DWDM Evolution

GEN5 – 80-102 GBaud | GEN6 - ~200 GBaud

System Based – Routers | OTN | Sleds



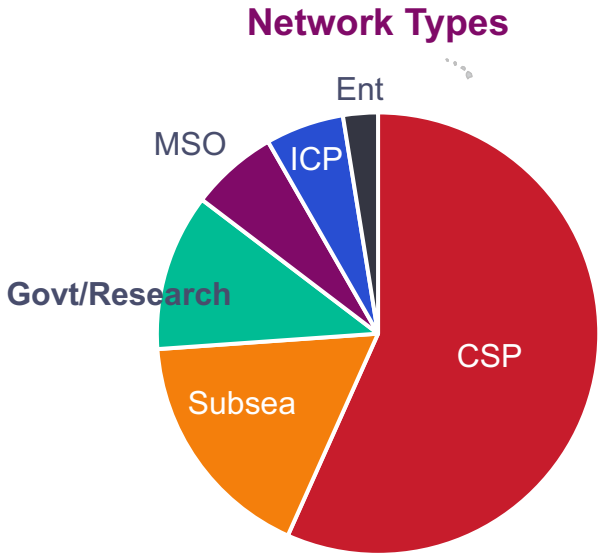
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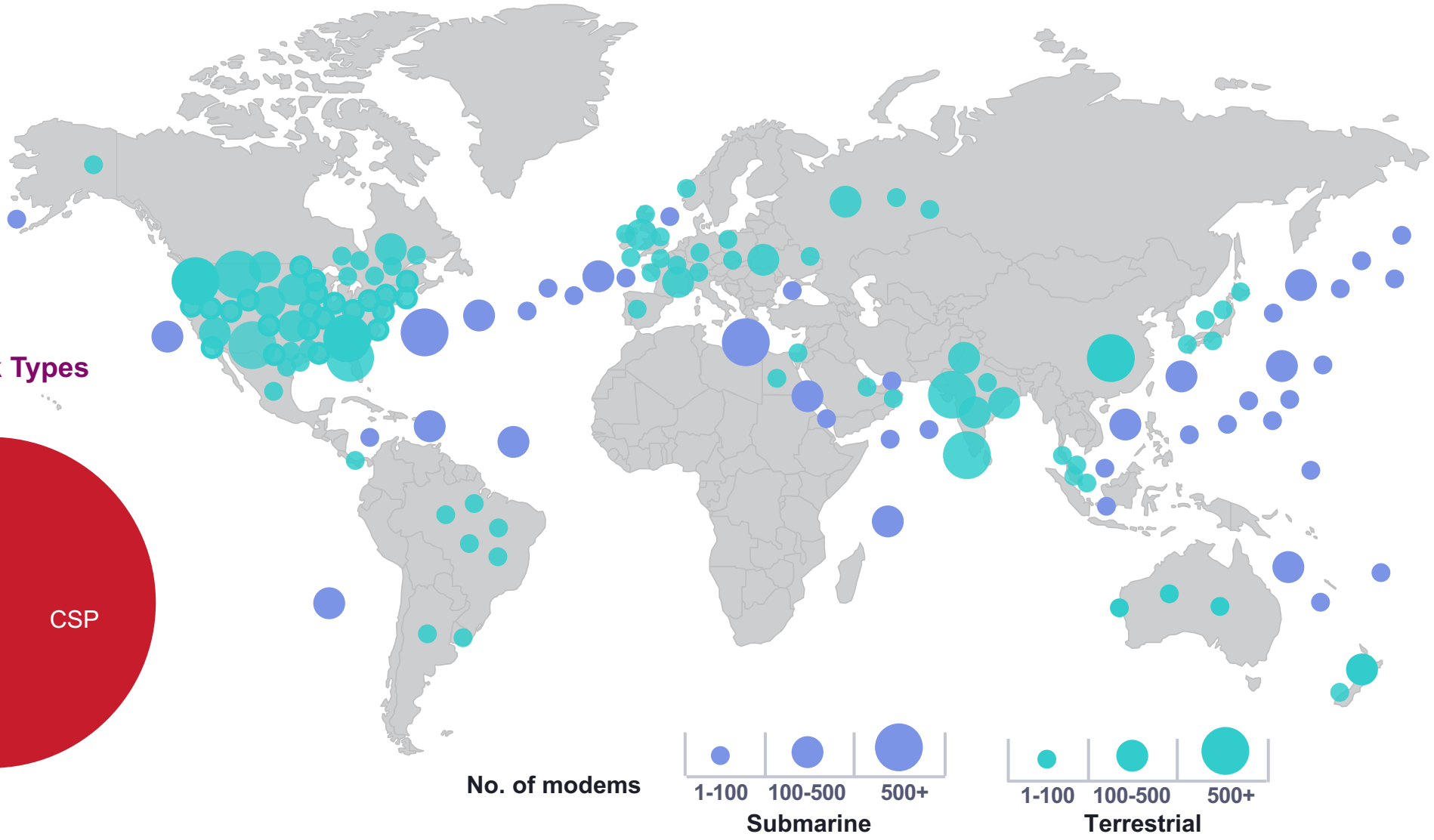
Strong Adoption of GEN5 800G technology

> 45,000 WL5e shipped, >185 customers

Benefits:

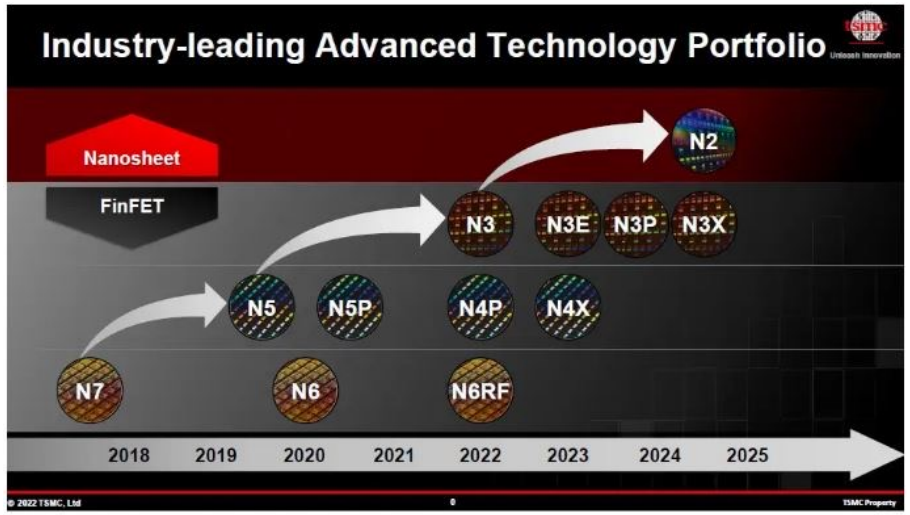
- 2 x capacity/wave
- 50% power
- 50% space





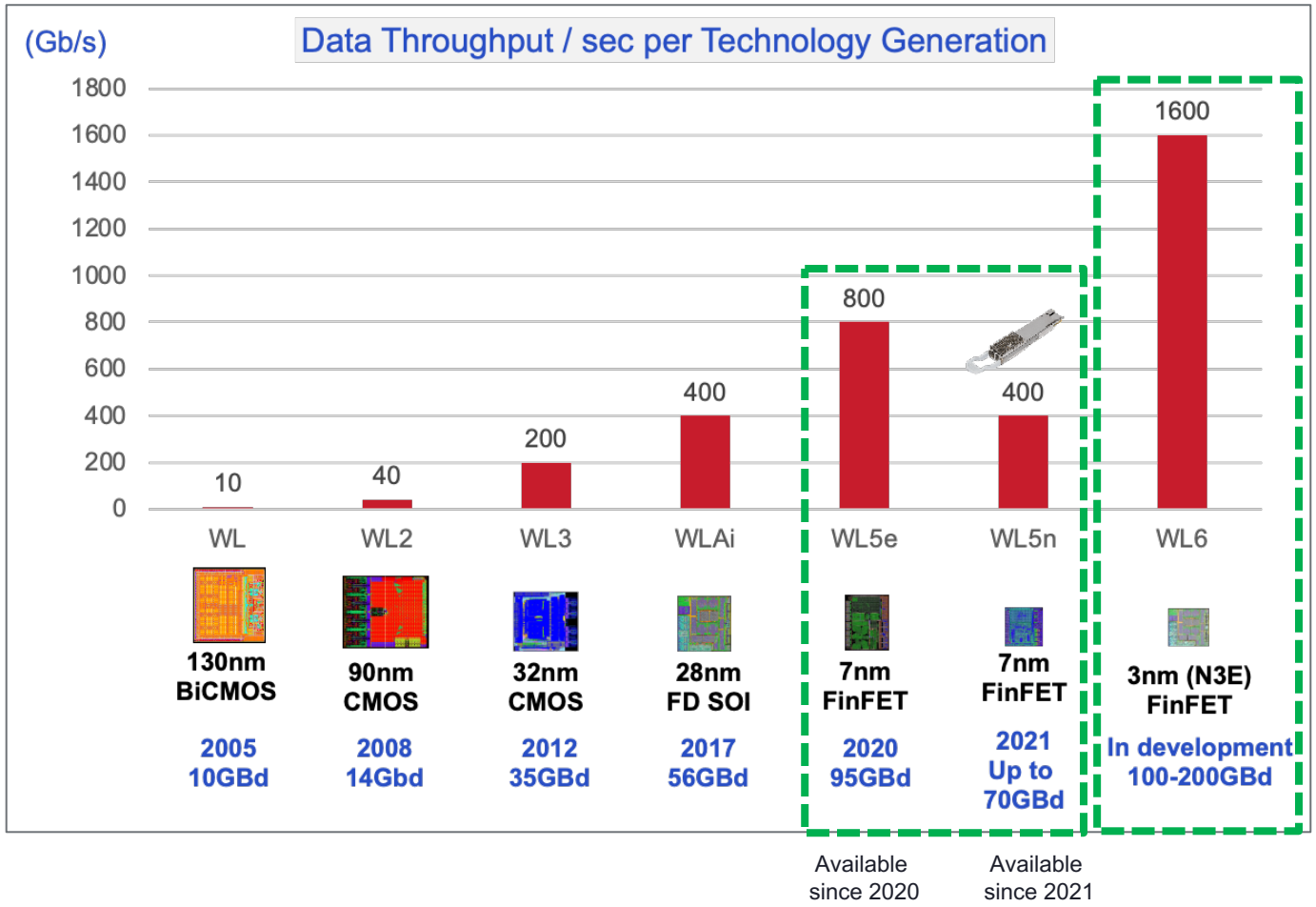
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Ciena – GEN6 Coherent Innovation



TSMC Europe Technology Symposium – Jun 20th, 2022

N3E PPA (vs. N5 V1.0)			
Speed Improvement at Same Power	Power Reduction at Same Speed	Logic Density	Chip Density*
15~20%	30~35%	~1.6x	~1.3X



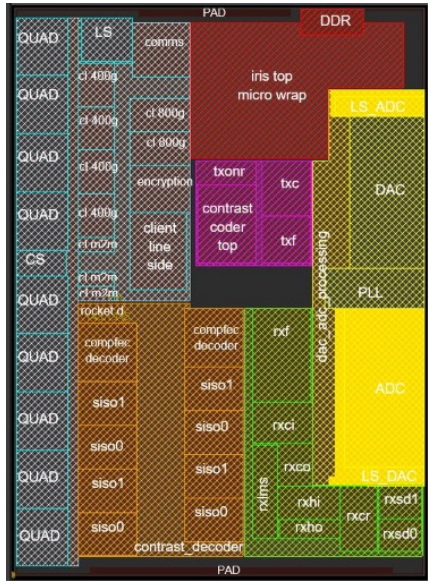
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Sustained forward innovation and investment to continue delivering industry leading solutions that address all networking requirements

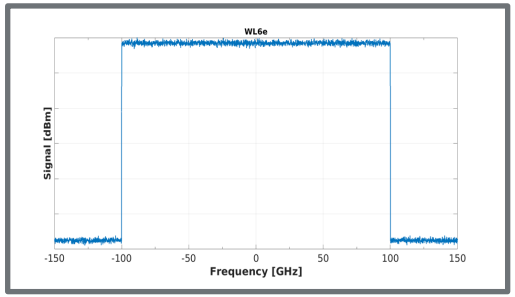


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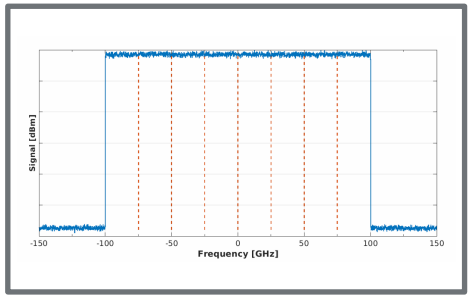
GEN6: DSP Improvements



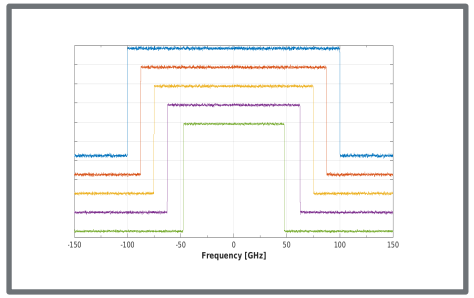
- 1 billion gates
- 1600 T ops
- 11x15 mm



Edgeless clock recovery



Variable FDM (from 1 to 8)

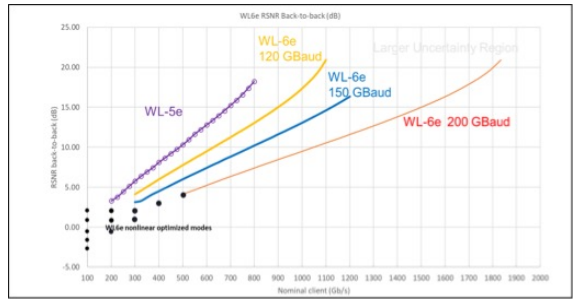


Variable Baud

WL6e DSP perfects Tx shaping with **Edgeless Clock Recovery**

- → Completely rectangular spectra
- → Further improvement in spectral efficiency

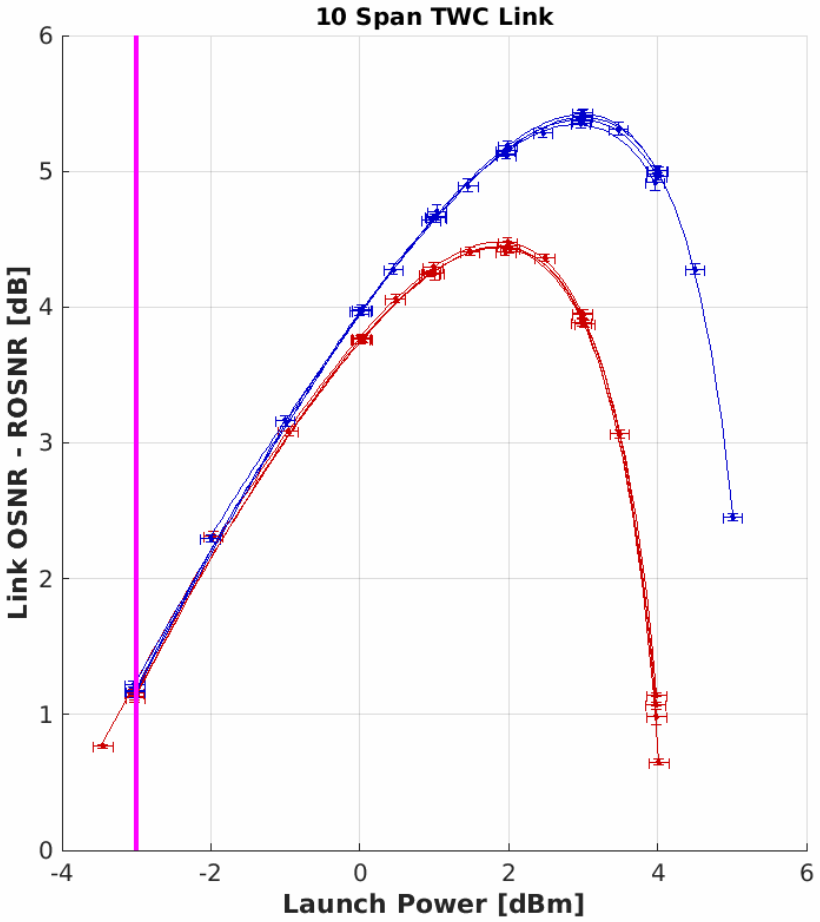
Mode	FDM setting typical for this total Dispersion, ns/nm		Amount of compensation for Chromatic Dispersion, ns/nm	
	Min	Max	Tx	Rx
1-FDM	-1	5	-20 to 20	-20 to 20
2-FDM	2	20	-40 to 40	-40 to 40
4-FDM	10	100	-40 to 80	-80 to 80
8-FDM	40	400	-40 to 125	-125 to 125

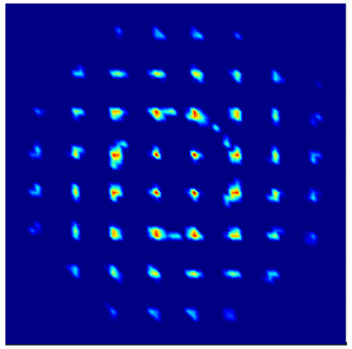


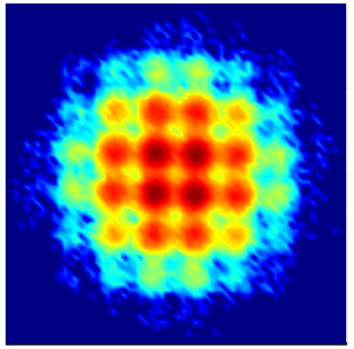
Enabling the Next Generation of Scalable and Programmable Adaptive Networks

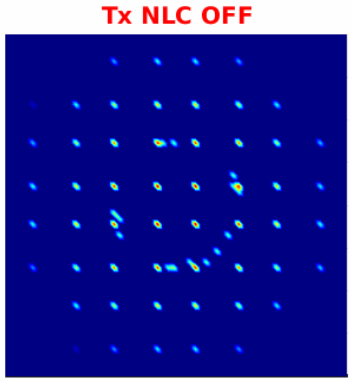
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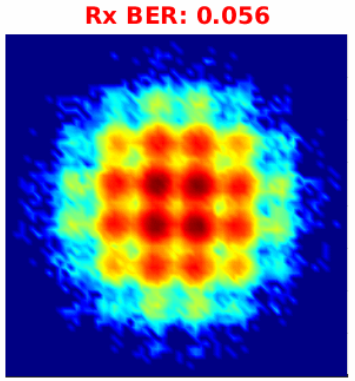
GEN 6 Extreme DSP Innovation: Nonlinear compensation

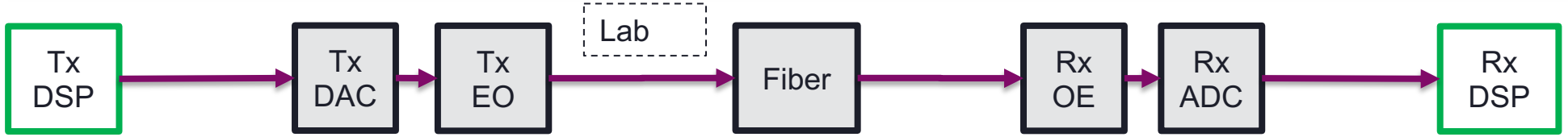












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Thank You

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